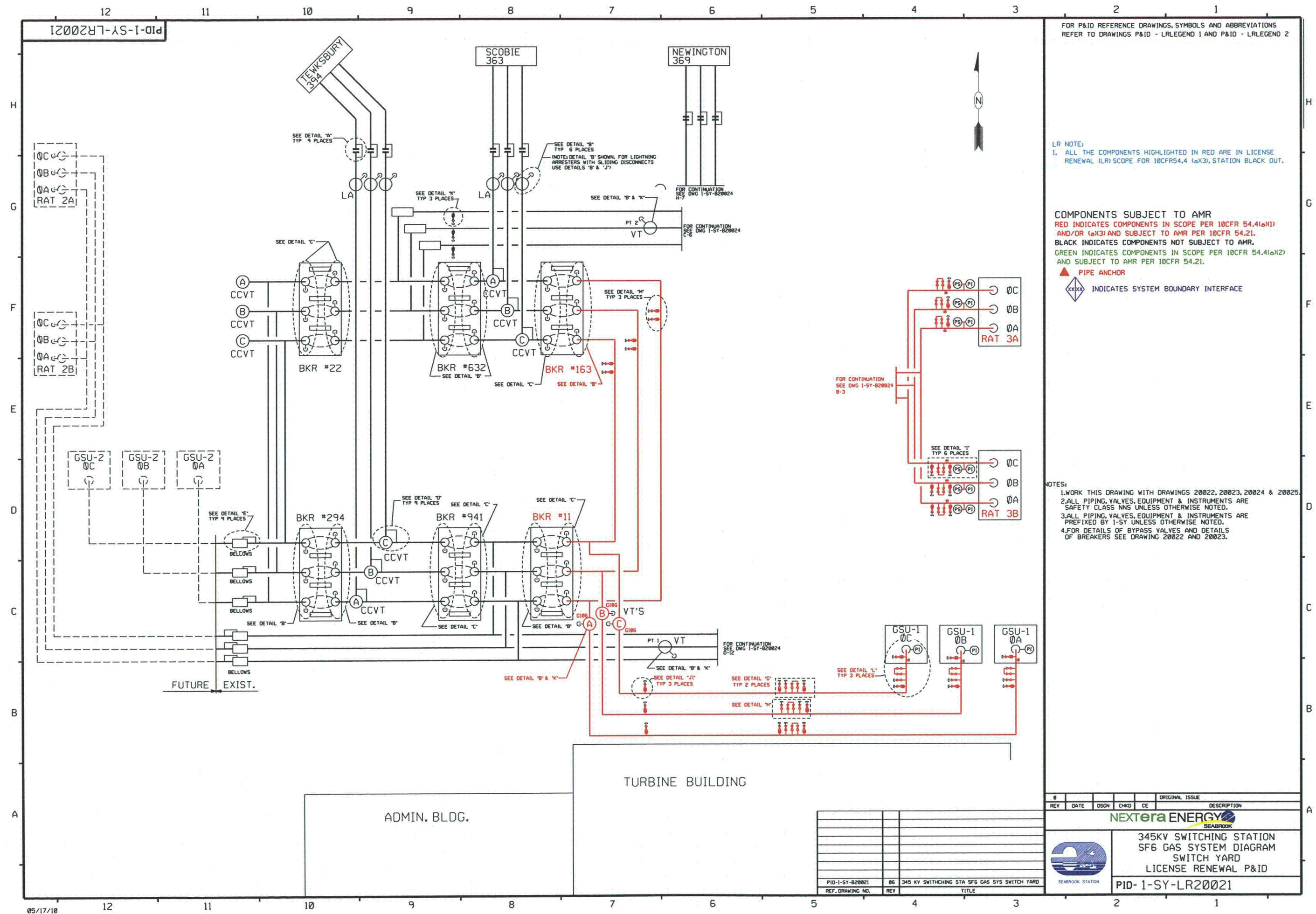






PID-1-SY-LR20023

FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS
REFER TO DRAWINGS P&ID - LRLEGEND 1 AND P&ID - LRLEGEND 2

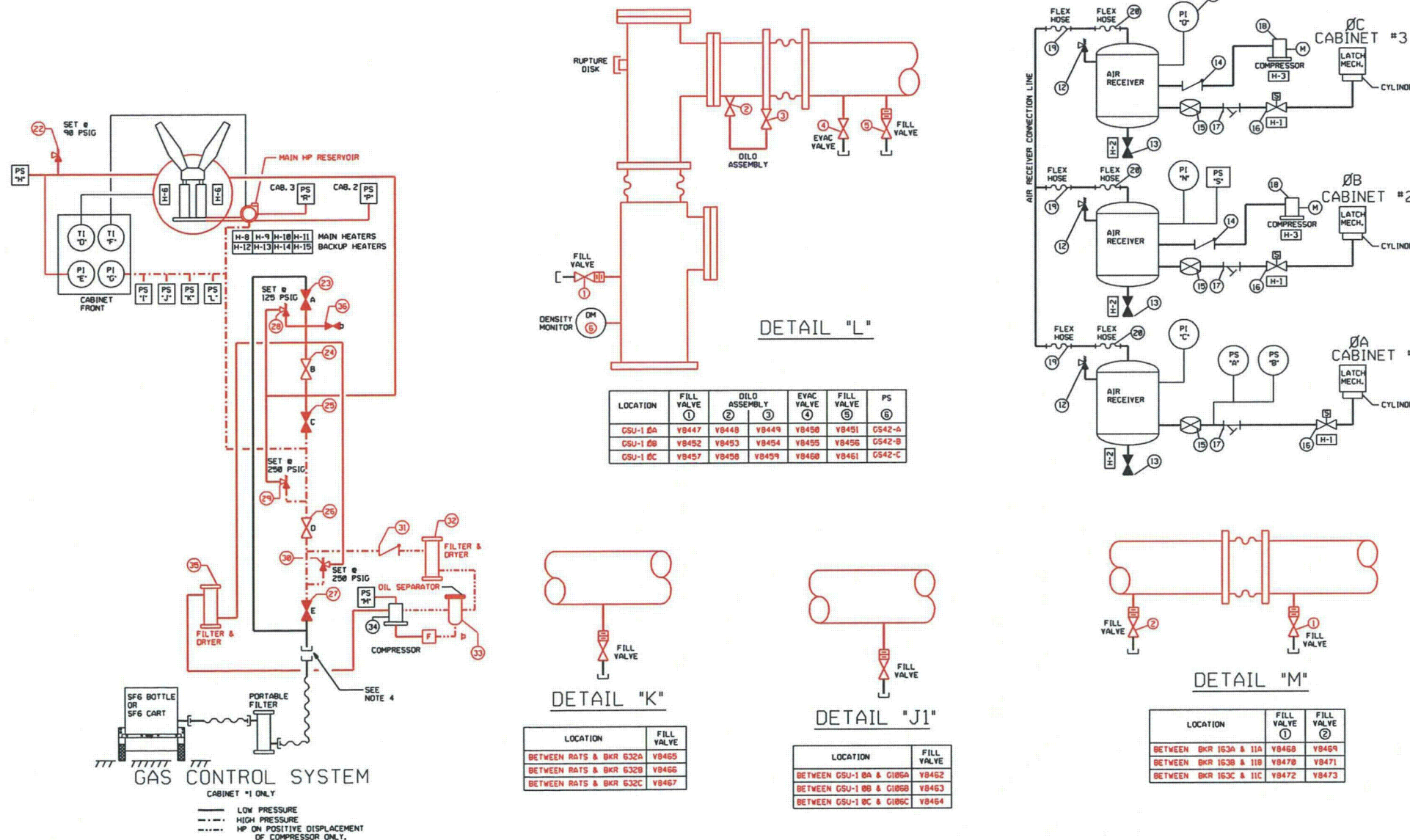
PNEUMATIC OPERATING SYSTEM																																
LOCATION	CABINET '1'										CABINET '2'										CABINET '3'											
	SAFETY VALVE 12	DRAIN VALVE 13	SHUTOFF VALVE 15	SOLENOID VALVE 16	STRAINER 17	FLEX HOSE 18	FLEX HOSE 20	PS 'A'	PS 'B'	PI 'C'	SAFETY VALVE 12	DRAIN VALVE 13	CHECK VALVE 14	SHUTOFF VALVE 15	SOLENOID VALVE 16	STRAINER 17	COMPRESSOR 18	FLEX HOSE 19	FLEX HOSE 20	PI 'N'	PS 'S'	SAFETY VALVE 12	DRAIN VALVE 13	CHECK VALVE 14	SHUTOFF VALVE 15	SOLENOID VALVE 16	STRAINER 17	COMPRESSOR 18	FLEX HOSE 19	FLEX HOSE 20	PI 'O'	
BREAKER *22	Y8220	Y8219	Y8221	Y8222	S-8000	MM-8038	MM-8062	PS-8030Z	PS-8029Z	PI-8023Z	Y8225	Y8225	Y8403	Y8223	Y8224	S-8001	C-8000	MM-8039	MM-8063	PI-8024Z	PS-8028Z	Y8230	Y8229	Y8520	Y8227	Y8228	S-8002	C-8024	MM-8048	MM-8064	PI-8025Z	
BREAKER *632	Y8245	Y8244	Y8242	Y8243	S-8003	MM-8041	MM-8065	PS-8040Z	PS-8047Z	PI-8041Z	Y8248	Y8249	Y8404	Y8246	Y8247	S-8004	C-8001	MM-8042	MM-8066	PI-8042Z	PS-8046Z	Y8252	Y8253	Y8521	Y8250	Y8251	S-8005	C-8025	MM-8043	MM-8067	PI-8043Z	
BREAKER *163	Y8266	Y8267	Y8264	Y8265	S-8006	MM-8044	MM-8068	PS-8047Z	PS-8046Z	PI-8040Z	Y8270	Y8271	Y8405	Y8268	Y8269	S-8007	C-8002	MM-8045	MM-8069	PI-8041Z	PS-8045Z	Y8276	Y8277	Y8522	Y8274	Y8275	S-8008	C-8026	MM-8046	MM-8070	PI-8042Z	
BREAKER *294	Y8308	Y8309	Y8306	Y8307	S-8012	MM-8050	MM-8074	PS-8081Z	PS-8080Z	PI-8074Z	Y8392	Y8393	Y8407	Y8390	Y8391	S-8013	C-8004	MM-8051	MM-8075	PI-8075Z	PS-8079Z	Y8396	Y8397	Y8524	Y8394	Y8395	S-8014	C-8028	MM-8052	MM-8076	PI-8076Z	
BREAKER *941	Y8364	Y8365	Y8362	Y8363	S-8015	MM-8053	MM-8077	PS-8115Z	PS-8114Z	PI-8108Z	Y8369	Y8368	Y8408	Y8366	Y8367	S-8016	C-8005	MM-8054	MM-8078	PI-8109Z	PS-8113Z	Y8372	Y8373	Y8525	Y8370	Y8371	S-8017	C-8029	MM-8055	MM-8079	PI-8110Z	
BREAKER *11	Y8341	Y8342	Y8339	Y8340	S-8018	MM-8056	MM-8080	PS-8015Z	PS-8014Z	PI-8008Z	Y8345	Y8346	Y8409	Y8343	Y8344	S-8019	C-8006	MM-8057	MM-8081	PI-8009Z	PS-8013Z	Y8349	Y8350	Y8526	Y8347	Y8348	S-8020	C-8030	MM-8058	MM-8082	PI-8010Z	

LR NOTE:
1. ALL THE COMPONENTS HIGHLIGHTED IN RED ARE IN LICENSE RENEWAL (LR) SCOPE FOR 10CFR54.4 (a)(3), STATION BLACK OUT.

COMPONENTS SUBJECT TO AMR
RED INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(1) AND/OR (a)(3) AND SUBJECT TO AMR PER 10CFR 54.21.
BLACK INDICATES COMPONENTS NOT SUBJECT TO AMR.
GREEN INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(2) AND SUBJECT TO AMR PER 10CFR 54.21.

▲ PIPE ANCHOR
◇ INDICATES SYSTEM BOUNDARY INTERFACE

NOTES:
1. WORK THIS DRAWING WITH DRAWING 20021, 20022, 20024 AND 20025.
2. ALL PIPING, VALVES, EQUIPMENT AND INSTRUMENTS ARE SAFETY CLASS UNLESS NOTED OTHERWISE NOTED.
3. ALL PIPING, VALVES, EQUIPMENT AND INSTRUMENTS ARE PREFIXED BY 1-SY UNLESS NOTED OTHERWISE NOTED.
4. LOCATIONS NOTED WITH A # HAVE HAD THE CAP REPLACED WITH A DILO FITTING PART # 3-401-R001 (CID# 431653 OR SIMILAR)

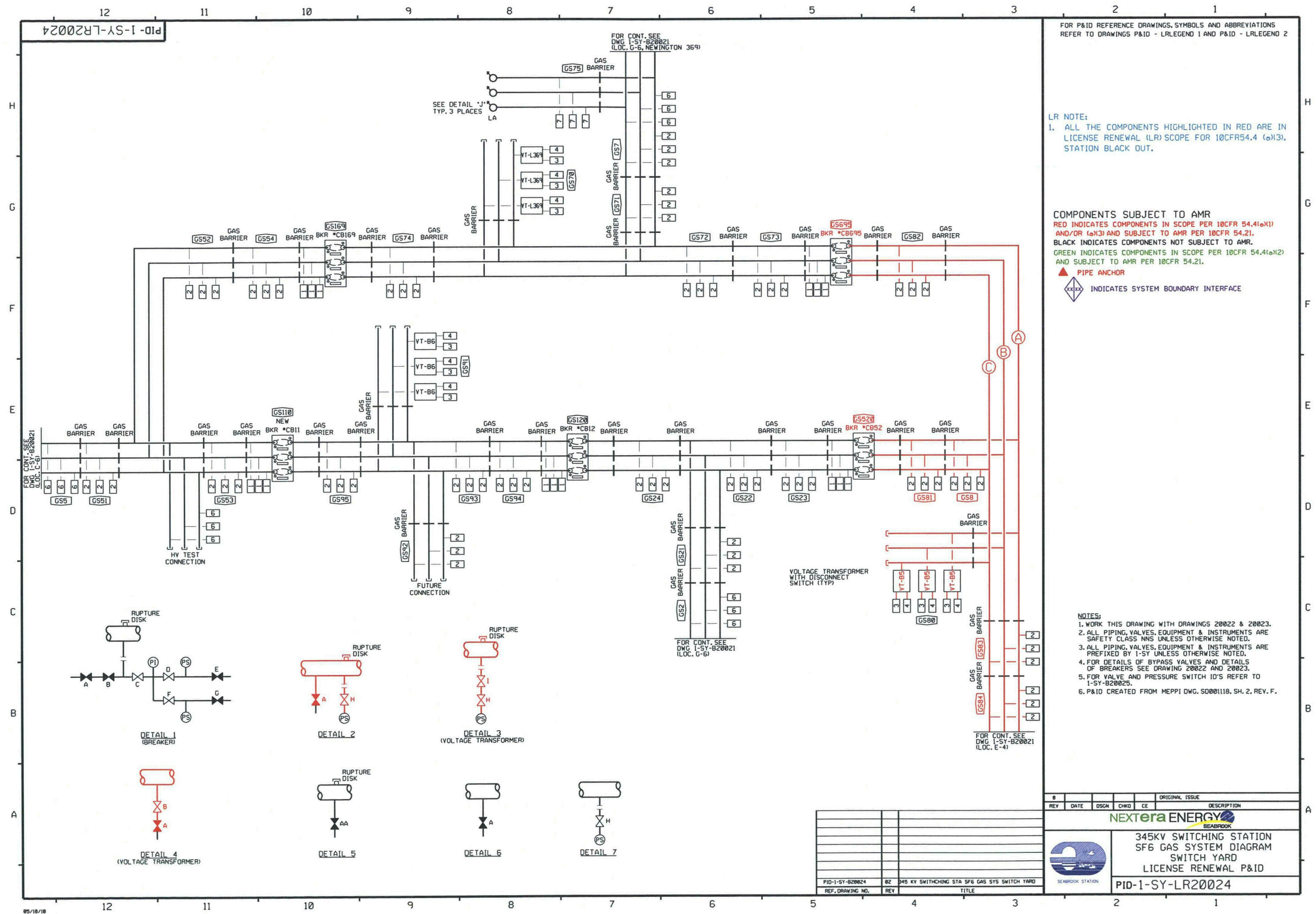


GAS CONTROL SYSTEM																													
LOCATION		CABINET #1 ONLY																											
	SAFETY VALVE 22	VALVE A 23	VALVE B 24	VALVE C 25	VALVE D 26	VALVE E 27	RELIEF 28	RELIEF 29	RELIEF 30	CHECK VALVE 31	FILTER DRYER 32	OIL SEP. 33	COMP. 34	FILTER DRYER 35	SAMPLE VALVE 36	PS "H"	TI "D"	TI "F"	PI "E"	PI "G"	PS "I"	PS "J"	PS "K"	PS "L"	PS "M"	PS "N"	SEE NOTE		
BREAKER #22	Y8218	Y8213	Y8212	Y8211	Y8218	Y8209	Y8215	Y8216	Y8217	Y8208	F-8000	MM-8000	C-8015	F-8009	Y8214	PS-8032Z	TI-8039Z	TI-8038Z	PI-8027Z	PI-8025Z	PS-8037Z	PS-8036Z	PS-8035Z	PS-8031Z	PS-8034Z	PS-8033Z	PS-8143Z		
BREAKER #532	Y8248	Y8235	Y8234	Y8233	Y8232	Y8231	Y8237	Y8238	Y8239	Y8241	F-8001	MM-8001	C-8017	F-8009	Y8236	PS-8100Z	TI-8107Z	TI-8106Z	PI-8095Z	PI-8094Z	PS-8105Z	PS-8104Z	PS-8103Z	PS-8099Z	PS-8102Z	PS-8101Z	PS-8147Z		
BREAKER #163	Y8263	Y8258	Y8257	Y8256	Y8255	Y8254	Y8259	Y8260	Y8261	Y8262	F-8002	MM-8002	C-8018	F-8010	Y8399	PS-8049Z	TI-8056Z	TI-8055Z	PI-8044Z	PI-8043Z	PS-8054Z	PS-8053Z	PS-8052Z	PS-8048Z	PS-8050Z	PS-8051Z	PS-8144Z		
BREAKER #294	Y8305	Y8309	Y8308	Y8307	Y8306	Y8305	Y8308	Y8309	Y8302	Y8303	F-8004	MM-8004	C-8020	F-8012	Y8304	PS-8083Z	TI-8080Z	TI-8080Z	PI-8078Z	PI-8077Z	PS-8088Z	PS-8087Z	PS-8086Z	PS-8082Z	PS-8084Z	PS-8081Z	PS-8146Z		
BREAKER #941	Y8361	Y8355	Y8354	Y8353	Y8352	Y8351	Y8356	Y8357	Y8358	Y8360	F-8005	MM-8005	C-8021	F-8013	Y8360	PS-8117Z	TI-8124Z	TI-8123Z	PI-8112Z	PI-8111Z	PS-8122Z	PS-8121Z	PS-8120Z	PS-8116Z	PS-8119Z	PS-8118Z	PS-8149Z		
BREAKER #11	Y8337	Y8332	Y8331	Y8330	Y8329	Y8328	Y8333	Y8334	Y8335	Y8336	F-8006	MM-8006	C-8022	F-8014	Y8330	PS-8017Z	TI-8012Z	TI-8011Z	PI-8012Z	PI-8011Z	PS-8022Z	PS-8021Z	PS-8020Z	PS-8016Z	PS-8019Z	PS-8018Z	PS-8142Z		

REV	DATE	DSGN	CHKD	CE	DESCRIPTION
1					ORIGINAL ISSUE
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					

PID-1-SY-020023
REF. DRAWING NO. 6
REV. 6
TITLE 345KV SWITCHING STATION SF6 GAS SYS DIAG DET

345KV SWITCHING STATION
SF6 GAS SYSTEM DIAGRAM
DETAILS
LICENSE RENEWAL P&ID
PID-1-SY-LR20023



PID-1-SY-LR20025

FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS
REFER TO DRAWINGS P&ID - LRLEGENO 1 AND P&ID - LRLEGENO 2

GAS ZONE	GAS COMPARTMENT	VALVE A	VALVE AA	VALVE B	VALVE C	VALVE D	VALVE E	VALVE F	VALVE G	VALVE H	VALVE I	PRESS SW 1	PRESS SW 2	PRESS SW 3	PRESS INDICATOR
GS120	GS120 PHASE A	1-SY-V-8528		1-SY-V-8529	1-SY-V-8530	1-SY-V-8531	1-SY-V-8532	1-SY-V-8533	1-SY-V-8534			1-SY-PS-GS120-A-1	1-SY-PS-GS120-A-2		1-SY-PI-GS120-A
	GS120 PHASE B	1-SY-V-8620		1-SY-V-8621	1-SY-V-8622	1-SY-V-8623	1-SY-V-8624	1-SY-V-8625	1-SY-V-8626			1-SY-PS-GS120-B-1	1-SY-PS-GS120-B-2		1-SY-PI-GS120-B
	GS120 PHASE C	1-SY-V-8712		1-SY-V-8713	1-SY-V-8714	1-SY-V-8715	1-SY-V-8716	1-SY-V-8717	1-SY-V-8718			1-SY-PS-GS120-C-1	1-SY-PS-GS120-C-2		1-SY-PI-GS120-C
GS24	GS24 PHASE A	1-SY-V-8535								1-SY-V-8536				1-SY-PS-GS24-A	
	GS24 PHASE B	1-SY-V-8627								1-SY-V-8628				1-SY-PS-GS24-B	
	GS24 PHASE C	1-SY-V-8719								1-SY-V-8720				1-SY-PS-GS24-C	
GS22	GS22 PHASE A	1-SY-V-8537								1-SY-V-8538				1-SY-PS-GS22-A	
	GS22 PHASE B	1-SY-V-8629								1-SY-V-8630				1-SY-PS-GS22-B	
	GS22 PHASE C	1-SY-V-8721								1-SY-V-8722				1-SY-PS-GS22-C	
GS23	GS23 PHASE A	1-SY-V-8539								1-SY-V-8540				1-SY-PS-GS23-A	
	GS23 PHASE B	1-SY-V-8631								1-SY-V-8632				1-SY-PS-GS23-B	
	GS23 PHASE C	1-SY-V-8723								1-SY-V-8724				1-SY-PS-GS23-C	
GS21	GS21 PHASE A	1-SY-V-8541								1-SY-V-8542				1-SY-PS-GS21-A	
	GS21 PHASE B	1-SY-V-8633								1-SY-V-8634				1-SY-PS-GS21-B	
	GS21 PHASE C	1-SY-V-8725								1-SY-V-8726				1-SY-PS-GS21-C	
GS520	GS520 PHASE A	1-SY-V-8543		1-SY-V-8544	1-SY-V-8545	1-SY-V-8546	1-SY-V-8547	1-SY-V-8548	1-SY-V-8549			1-SY-PS-GS520-A-1	1-SY-PS-GS520-A-2		1-SY-PI-GS520-A
	GS520 PHASE B	1-SY-V-8635		1-SY-V-8636	1-SY-V-8637	1-SY-V-8638	1-SY-V-8639	1-SY-V-8640	1-SY-V-8641			1-SY-PS-GS520-B-1	1-SY-PS-GS520-B-2		1-SY-PI-GS520-B
	GS520 PHASE C	1-SY-V-8727		1-SY-V-8728	1-SY-V-8729	1-SY-V-8730	1-SY-V-8731	1-SY-V-8732	1-SY-V-8733			1-SY-PS-GS520-C-1	1-SY-PS-GS520-C-2		1-SY-PI-GS520-C
GS81	GS81 PHASE A	1-SY-V-8550								1-SY-V-8551				1-SY-PS-GS81-A	
	GS81 PHASE B	1-SY-V-8642								1-SY-V-8643				1-SY-PS-GS81-B	
	GS81 PHASE C	1-SY-V-8734								1-SY-V-8735				1-SY-PS-GS81-C	
GS8	GS8 PHASE A	1-SY-V-8552								1-SY-V-8553				1-SY-PS-GS8-A	
	GS8 PHASE B	1-SY-V-8644								1-SY-V-8645				1-SY-PS-GS8-B	
	GS8 PHASE C	1-SY-V-8736								1-SY-V-8737				1-SY-PS-GS8-C	
GS80	GS80 PHASE A	1-SY-V-8554		1-SY-V-8804						1-SY-V-8555	1-SY-V-8805			1-SY-PS-GS80-A	
	GS80 PHASE B	1-SY-V-8646		1-SY-V-8806						1-SY-V-8647	1-SY-V-8807			1-SY-PS-GS80-B	
	GS80 PHASE C	1-SY-V-8738		1-SY-V-8808						1-SY-V-8739	1-SY-V-8809			1-SY-PS-GS80-C	
GS83	GS83 PHASE A	1-SY-V-8556								1-SY-V-8557				1-SY-PS-GS83-A	
	GS83 PHASE B	1-SY-V-8648								1-SY-V-8649				1-SY-PS-GS83-B	
	GS83 PHASE C	1-SY-V-8740								1-SY-V-8741				1-SY-PS-GS83-C	
GS82	GS82 PHASE A	1-SY-V-8558								1-SY-V-8559				1-SY-PS-GS82-A	
	GS82 PHASE B	1-SY-V-8650								1-SY-V-8651				1-SY-PS-GS82-B	
	GS82 PHASE C	1-SY-V-8742								1-SY-V-8743				1-SY-PS-GS82-C	
GS695	GS695 PHASE A	1-SY-V-8560		1-SY-V-8561	1-SY-V-8562	1-SY-V-8563	1-SY-V-8564	1-SY-V-8565	1-SY-V-8566			1-SY-PS-GS695-A-1	1-SY-PS-GS695-A-2		1-SY-PI-GS695-A
	GS695 PHASE B	1-SY-V-8652		1-SY-V-8653	1-SY-V-8654	1-SY-V-8655	1-SY-V-8656	1-SY-V-8657	1-SY-V-8658			1-SY-PS-GS695-B-1	1-SY-PS-GS695-B-2		1-SY-PI-GS695-B
	GS695 PHASE C	1-SY-V-8744		1-SY-V-8745	1-SY-V-8746	1-SY-V-8747	1-SY-V-8748	1-SY-V-8749	1-SY-V-8750			1-SY-PS-GS695-C-1	1-SY-PS-GS695-C-2		1-SY-PI-GS695-C
GS73	GS73 PHASE A	1-SY-V-8567								1-SY-V-8568				1-SY-PS-GS73-A	
	GS73 PHASE B	1-SY-V-8659								1-SY-V-8660				1-SY-PS-GS73-B	
	GS73 PHASE C	1-SY-V-8751								1-SY-V-8752				1-SY-PS-GS73-C	
GS72	GS72 PHASE A	1-SY-V-8569								1-SY-V-8570				1-SY-PS-GS72-A	
	GS72 PHASE B	1-SY-V-8661								1-SY-V-8662				1-SY-PS-GS72-B	
	GS72 PHASE C	1-SY-V-8753								1-SY-V-8754				1-SY-PS-GS72-C	
GS70	GS70 PHASE A	1-SY-V-8571		1-SY-V-8810						1-SY-V-8572	1-SY-V-8811			1-SY-PS-GS70-A	
	GS70 PHASE B	1-SY-V-8663		1-SY-V-8812						1-SY-V-8664	1-SY-V-8813			1-SY-PS-GS70-B	
	GS70 PHASE C	1-SY-V-8755		1-SY-V-8814						1-SY-V-8756	1-SY-V-8815			1-SY-PS-GS70-C	
GS71	GS71 PHASE A	1-SY-V-8573								1-SY-V-8574				1-SY-PS-GS71-A	
	GS71 PHASE B	1-SY-V-8665								1-SY-V-8666				1-SY-PS-GS71-B	
	GS71 PHASE C	1-SY-V-8757								1-SY-V-8758				1-SY-PS-GS71-C	
GS74	GS74 PHASE A	1-SY-V-8575								1-SY-V-8576				1-SY-PS-GS74-A	
	GS74 PHASE B	1-SY-V-8667								1-SY-V-8668				1-SY-PS-GS74-B	
	GS74 PHASE C	1-SY-V-8759								1-SY-V-8760				1-SY-PS-GS74-C	
GS169	GS169 PHASE A	1-SY-V-8577		1-SY-V-8578	1-SY-V-8579	1-SY-V-8580	1-SY-V-8581	1-SY-V-8582	1-SY-V-8583			1-SY-PS-GS169-A-1	1-SY-PS-GS169-A-2		1-SY-PI-GS169-A
	GS169 PHASE B	1-SY-V-8669		1-SY-V-8670	1-SY-V-8671	1-SY-V-8672	1-SY-V-8673	1-SY-V-8674	1-SY-V-8675			1-SY-PS-GS169-B-1	1-SY-PS-GS169-B-2		1-SY-PI-GS169-B
	GS169 PHASE C	1-SY-V-8761		1-SY-V-8762	1-SY-V-8763	1-SY-V-8764	1-SY-V-8765	1-SY-V-8766	1-SY-V-8767			1-SY-PS-GS169-C-1	1-SY-PS-GS169-C-2		1-SY-PI-GS169-C
GS54	GS54 PHASE A	1-SY-V-8584								1-SY-V-8585				1-SY-PS-GS54-A	
	GS54 PHASE B	1-SY-V-8676								1-SY-V-8677				1-SY-PS-GS54-B	
	GS54 PHASE C	1-SY-V-8768								1-SY-V-8769				1-SY-PS-GS54-C	
GS52	GS52 PHASE A	1-SY-V-8582								1-SY-V-8587				1-SY-PS-GS52-A	
	GS52 PHASE B	1-SY-V-8673								1-SY-V-8679				1-SY-PS-GS52-B	
	GS52 PHASE C	1-SY-V-8770								1-SY-V-8771				1-SY-PS-GS52-C	
GS51	GS51 PHASE A	1-SY-V-8588								1-SY-V-8589				1-SY-PS-GS51-A	
	GS51 PHASE B	1-SY-V-8680								1-SY-V-8681				1-SY-PS-GS51-B	
	GS51 PHASE C	1-SY-V-8772								1-SY-V-8773				1-SY-PS-GS51-C	
GS53	GS53 PHASE A	1-SY-V-8590								1-SY-V-8591				1-SY-PS-GS53-A	
	GS53 PHASE B	1-SY-V-8682								1-SY-V-8683				1-SY-PS-GS53-B	
	GS53 PHASE C	1-SY-V-8774								1-SY-V-8775				1-SY-PS-GS53-C	
GS110	GS110 PHASE A	1-SY-V-8592		1-SY-V-8593	1-SY-V-8594	1-SY-V-8595	1-SY-V-8596	1-SY-V-8597	1-SY-V-8598			1-SY-PS-GS110-A-1	1-SY-PS-GS110-A-2		1-SY-PI-GS110-A
	GS110 PHASE B	1-SY-V-8684		1-SY-V-8685	1-SY-V-8686	1-SY-V-8687	1-SY-V-8688	1-SY-V-8689	1-SY-V-8690			1-SY-PS-GS110-B-1	1-SY-PS-GS110-B-2		1-SY-PI-GS110-B
	GS110 PHASE C	1-SY-V-8776		1-SY-V-8777	1-SY-V-8778	1-SY-V-8779	1-SY-V-8780	1-SY-V-8781	1-SY-V-8782			1-SY-PS-GS110-C-1	1-SY-PS-GS110-C-2		1-SY-PI-GS110-C
GS95	GS95 PHASE A	1-SY-V-8599								1-SY-V-8600				1-SY-PS-GS95-A	
	GS95 PHASE B	1-SY-V-8691								1-SY-V-8692				1-SY-PS-GS95-B	
	GS95 PHASE C	1-SY-V-8783								1-SY-V-8784				1-SY-PS-GS95-C	
GS93	GS93 PHASE A	1-SY-V-8601								1-SY-V-8602				1-SY-PS-GS93-A	
	GS93 PHASE B	1-SY-V-8693								1-SY-V-8694				1-SY-PS-GS93-B	
	GS93 PHASE C	1-SY-V-8785								1-SY-V-8786				1-SY-PS-GS93-C	
GS91	GS91 PHASE A	1-SY-V-8603		1-SY-V-8816						1-SY-V-8604	1-SY-V-8817			1-SY-PS-GS91-A	
	GS91 PHASE B	1-SY-V-8695		1-SY-V-8818						1-SY-V-8696	1-SY-V-8819			1-SY-PS-GS91-B	
	GS91 PHASE C	1-SY-V-8787		1-SY-V-8820						1-SY-V-8788	1-SY-V-8821			1-SY-PS-GS91-C	
GS94	GS94 PHASE A	1-SY-V-8605								1-SY-V-8606				1-SY-PS-GS94-A	
	GS94 PHASE B	1-SY-V-8697								1-SY-V-8698				1-SY-PS-GS94-B	
	GS94 PHASE C	1-SY-V-8789								1-SY-V-8790				1-SY-PS-GS94-C	
GS92	GS92 PHASE A	1-SY-V-8607								1-SY-V-8608				1-SY-PS-GS92-A	
	GS92 PHASE B	1-SY-V-8699								1-SY-V-8700				1-SY-PS-GS92-B	
	GS92 PHASE C	1-SY-V-8791								1-SY-V-8792				1-SY-PS-GS92-C	
GS2	GS2 PHASE A	1-SY-V-8609													
	GS2 PHASE B	1-SY-V-8701													
	GS2 PHASE C	1-SY-V-8793													
GS84	GS84 PHASE A	1-SY-V-8610								1-SY-V-8611				1-SY-PS-GS84-A	
	GS84 PHASE B	1-SY-V-8702								1-SY-V-8703				1-SY-PS-GS84-B	
	GS84 PHASE C	1-SY-V-8794								1-SY-V-8795				1-SY-PS-GS84-C	
GS7	GS7 PHASE A	1-SY-V-8612								1-SY-V-8613					